and yellow, but not red. G can see caps of all colours other than orange. If E is wearing an indigo coloured cap, then the colour of the cap worn by F is:

(a) blue (b) violet
(c) red (d) orange

Ans. (c) : As per the question 7 person are standing in the queue.

A ⟶ Green
B ⟶ Blue
C ⟶ Yellow
D ⟶ Violet
E ⟶ Indigo
F ⟶
G ⟶ Orange

So cap worn by f will be red in colour.

136. A person travelled from one place to another at an average speed of 40 kilometres/hour and back to the original place at an average speed of 50 kilometres/hour. What is his average speed in kilometres/hour during the entire round-trip?

(a) 45
(b) $20\sqrt{5}$
(c) 400/9
(d) Impossible to find out unless the distance between the two places is known

Ans. (c): Let the distance be *x* km

$\text{Time} = \frac{\text{distance}}{\text{speed}}$

Total time taken for going and coming back

$= \frac{x}{40} + \frac{x}{50}$

$= \frac{5x+4x}{200} = \frac{9x}{200}$

Total distance be covered = 2*x*

Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

$= \frac{2x}{9x/200} = \frac{200\times 2}{9} = \frac{400}{9}$ km/hr

137. The value of:
(a – m) (b – m) ... (y – m) (z – m) is

(a) $m^{26} + am^{25} + abm^{24} + ...+$ a.b.c. ...z
(b) $m^{26} - am^{25} + abm^{24} + ...-$ a.b.c. ...z
(c) 0
(d) Indeterminate

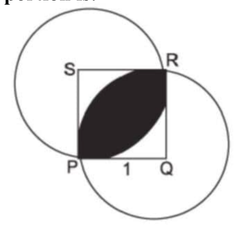
Ans. (c) : Value of (a – m) (d – m) ….. (y – m) (z – m)
= (a – m) × (b – m) ×…….(m – m) × (n – m) …. × (y – m) (z – m)
= m – m = 0
= 0 × {(a –m) (b – m) ….. (y – m) (z – m)}
= 0

138. A rectangular sump of dimensions 6 m x 5 m x 4 m is to be built by using bricks to make the outer dimension 6.2 m x 5.2 x 4.2 m. Approximately how many bricks of size 20 cm x 10 cm x 5 cm are required to build the sump for storing water?

(a) 1500 (b) 3000
(c) 15000 (d) 30000

Ans. (c) : Volume of rectangular jump = outer volume
Inner volume
$= (6.2 \times 5.2 \times 4.2) - (6 \times 5 \times 4)$
$= 15.408\ m^3$
Volume of 1 brick is $= 20 \times 10 \times 5 = 1000\ cm^3$
$= 0 : 001\ m^3$
So, number of bricks required is
$= \frac{15.408}{0.001} = 15408 \approx 15000$

139. Consider the figure given below:
PQRS is a square of side 1 unit and Q, S are the centres of the two circles. The area of the shaded portion is:



(a) $\frac{\pi}{2}$ (b) $\frac{1}{2}$
(c) $\frac{\pi}{4} - \frac{1}{2}$ (d) $\frac{\pi}{2} - 1$

Ans. (d) : The area of shaded portion is = (Area of curve RSP – Area of triangle PSR) × 2

$= \frac{1}{4}\pi(1)^2 - \frac{1}{2}\times 1\times 1$

$= \left(\frac{\pi}{4} - \frac{1}{2}\right)\times 2 = \frac{\pi}{2} - 1$

140. A thief running at 8 km/hr is chased by a policeman whose speed is 10 km/hr. If the thief is 100 metres ahead of the policeman, then the time required for the policeman to catch the thief will be:

(a) 2 minutes (b) 6 minutes
(c) 10 minutes (d) 3 minutes

Ans. (d) : Difference between speed of policeman and Thief is = (10 – 8) km/hr.
= 2km/hr
Time taken to cover 2 km is 1 hour
= 2000 meter in 60 minutes.
So, to cover 100 meter

$= \left(\frac{60}{2}\times\frac{100}{1000}\right)$ minutes
= 3 minutes

141. A student has 60% chance of passing in English and 54% chance of passing in both English and Mathematics. What is the percentage probability that he will fail in Mathematics?

(a) 12 (b) 36
(c) 4 (d) 10

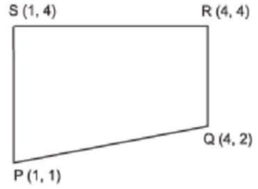
Ans. (d) : Ans. (d) : Chance of passing in English is 60% = 0.6
And chance for passing in both English and Mathematics is = 54% = 0.54
Percentage of Probability to pass in mathematics is
$= \frac{0.54}{6} = 0.9 = 90\%$
So percentage probability to fail in mathematics is
= 100 – 90 = 10%

142. A table has three drawers. It is known that one of the drawers contains two silver coins, another contains two gold coins and the third one contains a silver coin and a gold coin. One of the drawers is opened at random and a coin is drawn. It is found to be a silver coin. What is the probability that the other coin in the drawer is a gold coin?

(a) 0.25 (b) 1.00
(c) 0.50 (d) 0.60

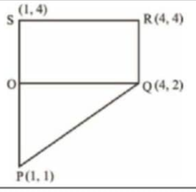
Ans. (c) : If one of the drawers is opened and a coin drawn is silver. The probability of getting gold coin in other two drawer is 0.5 or $\frac{1}{2}$.

143. In the Cartesian plane four points P, Q, R, S have co-ordinates (1, 1), (4, 2), (4, 4) and (1, 4). The area of the quadrilateral PQRS is:



(a) 9
(b) 7.5
(c) 4.5
(d) Impossible to find unless the lengths of the diagonals are known

Ans. (d) Ans. (b) :



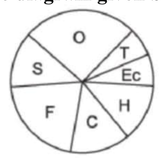
length of RQ = 4 – 2 = 2 unit
length of RS = 4 – 1 = 3 unit
So,
RS = OQ
OS = RQ
Area of Rectangle OSRQ = 2 × 3 = 6 unit

Area of Triangle $QOP = \frac{1}{2}(OQ) \times OP$

$= \frac{1}{2}\times 3 \times 1 = 1.5$ unit

{OP = SP – OS}
= (4 –1) unit 2 unit
= 3 – 2 = 1 unit
Thus total area = Area of Rectangle OS RQ + OQP
= 6 + 1.5 = 7.5 unit

144. Consider the diagram given below:



T : Transport
Ec : Education of children
H : Housing
C : Clothing
F : Food
S : Savings
O : Others

From the diagram shown it would be right to conclude that

(a) the family spent more than half the income on food and clothing
(b) the amount saved by the family was too little
(c) the family had no health problems
(d) the family managed to meet all the essential expenses out of the income earned

Ans. (d) : From the diagram shown it would be right to conclude that the family managed to meet all the essential expenses out of the income earned.

145. Consider the table given below providing details of traffic volume per hour for locations:

Location	Total traffic volume	% of Heavy Vehicles	Average noise level in dB (A)	Noise pollution level, in dB (A)
I	377	24.40	73.50	84.00
II	380	12.50	72.60	83.00
III	377	30.00	73.50	86.50
IV	225	12.50	72.98	80.90